Aman Sen

FPGA Design Engineer

RESUME_1863

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Total Experience: 4 years

PROFESSIONAL SUMMARY

FPGA Design Engineer with 4 years of experience in fpga, using VHDL, Verilog, Xilinx Vivado, Intel Quartus. Experienced in rtl simulation, fixed-point arithmetic and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: VHDL, Timing Closure, Intel Quartus, DSP, Python, Verilog, AXI4, FPGA, SPI, UART, Hardware Simulation, Xilinx Vivado

Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

PROFESSIONAL EXPERIENCE

FPGA Design Engineer | SparrowCloud Labs | Jan 2024 - Present

- Improved maximum operating frequency by 29% after timing-path optimization
- Verified 660 protocol corner cases with repeatable simulation suites
- Reduced FPGA resource utilization by 27% through arithmetic sharing
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Associate FPGA Design Engineer | KoraLink Innovations | Jul 2022 - Dec 2023

- Reduced FPGA resource utilization by 12% through arithmetic sharing
- Lowered end-to-end data-path latency by 41% using pipelined logic

PROJECTS

Protocol Adapter Core | AXI4, Verilog, FPGA, Intel Quartus

- Developed reusable SPI and UART modules with loopback tests.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

Camera Data Bridge | SPI, AXI4, DSP, VHDL

- Designed AXI streaming logic, frame buffers and clock-domain boundaries.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

EDUCATION

B.Tech in Electronics and Communication Engineering, Pragati Institute of Applied Sciences, 2022

Relevant Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

CERTIFICATIONS

Digital Signal Processing Basics (2026)